Chapter 5, Section 4

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1. The linear system of conics in $\mathbb{P}^2$ with two assigned base points P_1 and P_2 (4.1) determines a morphism ψ of X' (which is $\mathbb{P}^2$ with P_1 and P_2 blown up) to a nonsingular quadric surface Y in $\mathbb{P}^3$, and furthermore X' via ψ is isomorphic to Y with one point blown up.

Proof. We have the commutative diagram

$$\begin{array}{ccccc} & & X' & & \\ & \swarrow^{\pi} & & \searrow^{\psi} & \\ \mathbb{P}^2 & & \overset{\varphi}{\dashrightarrow} & & Y \hookrightarrow \mathbb{P}^3 \end{array},$$

where φ is defined by the linear system $\mathfrak{d}$ of conics on $\mathbb{P}^2$ with assigned base-points P_1 and P_2, Y is the image of φ, and X' is the blow-up of $\mathbb{P}^2$ at P_1 and P_2. We may take the two points to be $P_1 = (1:0:0)$ and $P_2 = (0:1:0)$. Then the vector space $V \subseteq \mathrm{H}^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2))$ corresponding to $\mathfrak{d}$ is spanned by

$$y_0 = x_1x_2, \quad y_1 = x_0x_2, \quad y_2 = x_0x_1, \quad y_3 = x_2^2,$$

where y_i are the homogenous coordinates of $\mathbb{P}^3$. In particular, φ is defined as

$$\varphi(x_0 : x_1 : x_2) = (y_0 : y_1 : y_2 : y_3),$$

and the points in the image of ψ satisfies the equation

$$y_0y_1 - y_2y_3 = 0,$$

which corresponds to a nonsingular quadric surface $Y \subset \mathbb{P}^3$. It is a birational map since if we restrict to the affine patch defined by $x_2 \neq 0$, then we can identify φ with a quadratic transformation (4.2.3).

For each $i = 1, 2$, let E_i be the strict transform of P_i, let M be the line through P_1 and P_2, which is defined by $x_2 = 0$ in $\mathbb{P}^2$, and let $\tilde{M}$ be the strict transform of M in X'. By (3.6), we have

$$\tilde{M} \sim \pi^*M - E_1 - E_2,$$

so by (3.2), $\tilde{M}$ has self intersection

$$\tilde{M}^2 = (M)^2 + (E_1)^2 + (E_2)^2 = -1.$$

Let λ be a local coordinate on M such that M consists of the points $(\lambda : 1 : 0)$. The image of M under φ for $\lambda \neq 0$ is computed to be

$$\varphi(\lambda : 1 : 0) = (0 : 0 : \lambda : 0),$$

so denote by Q the point $\varphi(M - \{P_1, P_2\}) = (0 : 0 : 1 : 0)$. We claim X' can be identified with the blowing-up of Y at Q. Indeed, φ induces an isomorphism

$$\mathbb{P}^2 - M \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and π induces an isomorphism

$$X' - E_1 \cup E_2 \cong \mathbb{P}^2 - \{P_1, P_2\},$$

so ψ induces an isomorphism

$$X' - \tilde{M} \cup E_1 \cup E_2 \cong Y - \{Q\} \cup \psi(E_1) \cup \psi(E_2),$$

and since $\psi(E_i)$ are lines in $\mathbb{P}^3$ which intersect at Q, we conclude that ψ induces an isomorphism

$$X' - \tilde{M} \cong Y - \{Q\}.$$

Hence by (5.7), X' can be identified with Y blown-up at Q. (See [Manin, §18] for more details.) □

2. Let φ be the quadratic transformation of (4.2.3), centered at P_1, P_2, P_3. If C is an irreducible curve of degree d in $\mathbb{P}^2$, with points of multiplicity r_1, r_2, r_3 at P_1, P_2, P_3, then the strict transform C' of C by φ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at Q_1, $d - r_1 - r_3$ at Q_2, and $d - r_1 - r_2$ at Q_3. The curve may have arbitrary singularities.

Proof. We have a commutative diagram

$$\begin{array}{ccc} & X & \\ {\scriptstyle\pi}\swarrow & & \searrow{\scriptstyle\psi} \\ \mathbb{P}^2 & \overset{\varphi}{\dashrightarrow} & \mathbb{P}^2 \end{array} .$$

Let H be a hyperplane section of $\mathbb{P}^2$, so that φ is given by the linear system $\mathfrak{d} = |2H - P_1 - P_2 - P_3|$, and let E_1, E_2, E_3 be the exceptional divisors of X via the blowing-up of $\mathbb{P}^2$ at P_1, P_2, P_3, respectively. By (3.6), the strict transform $\tilde{C}$ of C by π is linearly equivalent to

$$\tilde{C} = \pi^* C - r_1 E_1 - r_2 E_2 - r_3 E_3.$$

Since ψ is given by the linear system $\mathfrak{d}' = |D|$ with $D = 2\pi^* H - E_1 - E_2 - E_3$, φ is a birational map such that $C' = \varphi(C) = \psi(\tilde{C})$, and $C \sim dH$ in $\mathbb{P}^2$, the degree of C' in $\mathbb{P}^2$ can be computed using (Ex. 1.2) and (3.2)

$$\begin{aligned} \tilde{C}.D &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(2\pi^* H - E_1 - E_2 - E_3) \\ &= 2d(H^2) + r_1 (E_1)^2 + r_2 (E_2)^2 + r_3 (E_3)^2 \\ &= 2d - r_1 - r_2 - r_3. \end{aligned}$$

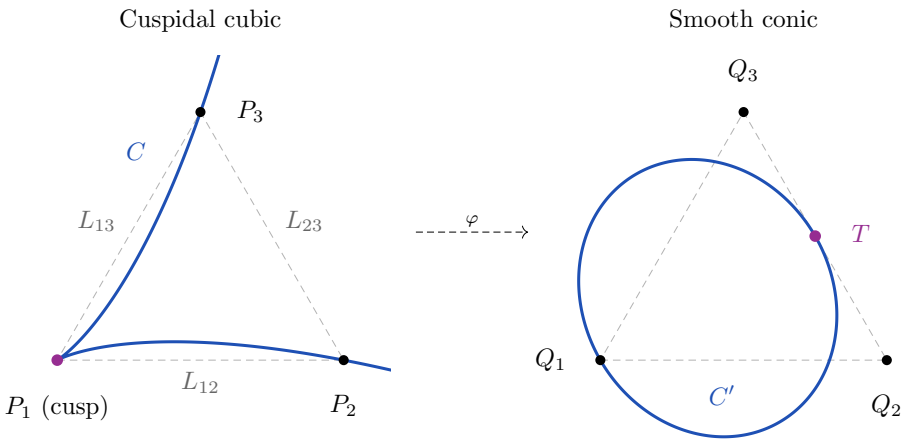
For each $i \neq j$, let L_{ij} be the line which goes through P_i and P_j. By (3.6), for each $k = 1, 2, 3$, the multiplicity of C' at Q_k is the intersection of $\tilde{C}$ with the strict transform $\tilde{L}_{ij}$ of L_{ij} with $i, j \neq k$ via π (4.2.3), which by (3.6) is linearly equivalent to

$$\tilde{L}_{ij} = \pi^* L_{ij} - E_i - E_j.$$

Again, we have $L_{ij} \sim H$ in $\mathbb{P}^2$, so

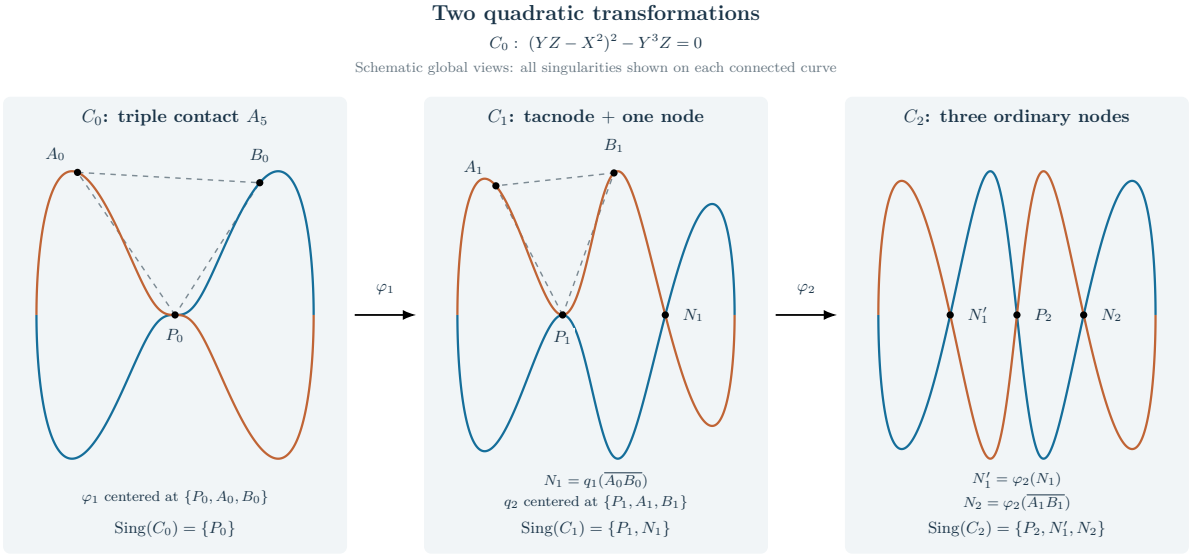
$$\begin{aligned} \tilde{C}.\tilde{L}_{ij} &= (d\pi^* H - r_1 E_1 - r_2 E_2 - r_3 E_3).(\pi^* H - E_i - E_j) \\ &= d(H)^2 + r_i (E_i)^2 + r_j (E_j)^2 \\ &= d - r_i - r_j. \end{aligned}$$

Hence, C' has points of multiplicity $d - r_i - r_j$ at Q_k. □



3. Let C be an irreducible curve in $\mathbb{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform C' of C has only *ordinary* singularities, i.e., multiple points with all distinct tangent directions.

Proof. If C is a conic in $\mathbb{P}^2$, then it is nonsingular, so assume C has degree ≥ 3, and let P_1 be a nonordinary singular point of C. Our goal is to find points $P_2, P_3 \in \mathbb{P}^2$ such that the quadratic transformation centered at P_i improves the singularity at P_1 and only introduces ordinary singularities on the strict transform of C. To that end, for each $i \neq j$, let L_{ij} be the line through P_i and P_j. The restriction of the quadratic transformation $\varphi : \mathbb{P}^2 \to \mathbb{P}^2$ centered at P_i to $\mathbb{P}^2 - \bigcup L_{ij}$ is an isomorphism onto its image, and each L_{ij} is sent to a point Q_k with $k \neq i, j$ such that if M is a curve meeting L_{ij} transversally away from the centers, then the strict transform of M has ordinary singularities at Q_k. Therefore, we require that away from P_1, the lines L_{ij} meets C transversally, and L_{12}, L_{13} are distinct from the tangent directions of C at P. There are only finitely many tangent directions of C at P_1, and since C is not strange (IV, 3.8) (IV, 3.9), we can find many lines L_{12} and L_{13} through P_1 with tangent vectors distinct from that of C at P_1 which meets C transversally. Lastly, pick L_{23} which avoids P_1, not parallel to L_{12}, L_{13}, meets C transversally, and the points P_2, P_3 of intersection of L_{23} with L_{12}, L_{13}, respectively, are not on C, which again exists because C is not strange. The strict transform of C under the quadratic transformation centered at P_i improves the singularity at P_1 (3.7) and only introduces ordinary singularities at Q_k. By (3.9), we can repeat this process for each nonordinary singularity until the strict transform only has ordinary singularities introduced by the quadratic transformations. □



4. (a) Use (4.5) to prove the following lemma on cubics: If C is an irreducible plane cubic curve, if L is a line meeting C in points P, Q, R, and L' is a line meeting C in points P', Q', R', let P'' be the third intersection of the line PP' with C, and define Q'', R'' similarly. Then P'', Q'', R'' are collinear.

(b) Let P_0 be an inflection point of C, and *define* the group operation on the set of regular points of C by the geometric recipe "let the line PQ meet C at R, and let P_0R meet C at T, then $P + Q = T$" as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

Proof.

(a) Let L_P be the line PP', and similarly for L_Q, L_R, and let M be the line going through P'' and Q''. The divisors C and $L_P + L_Q + L_R$ belong to the linear system of conics in $\mathbb{P}^2$ with assigned base points $P, P', P'', Q, Q'', Q''', R, R'$, and they intersect at R'', so by (4.5) R'' is the unique point in $\mathbb{P}^2$ such that every cubic curve in $\mathbb{P}^2$ going through $P, P', P'', Q, Q'', Q''', R, R'$ also goes through R''. In particular, $L + L' + M$ belongs to this linear system, so it must go through R''. Therefore, M must go through R''.

(b) Let P_1, P_2, P_3 be regular points of C, let L_{12} be the line P_1P_2, let R_{12} be the third point L_{12} meets C, let L'_{12} be the line P_0R_{12}, let $Q_{12} = P_1 + P_2$, and similarly for L_{23}, L'_{23}, and R_{23}, Q_{23}. Let $M_{12,3}$ be the line $Q_{12}P_3$, let $S_{12,3}$ be the third point $M_{12,3}$ meets C, let $M'_{12,3}$ be the line $P_0S_{12,3}$, let $T_{12,3} = Q_{12} + P_3$, and similarly for $M_{1,23}, M'_{1,23}$, and $S_{1,23}, T_{1,23}$. We want to show $T_{12,3} = T_{1,23}$. Applying (a) to

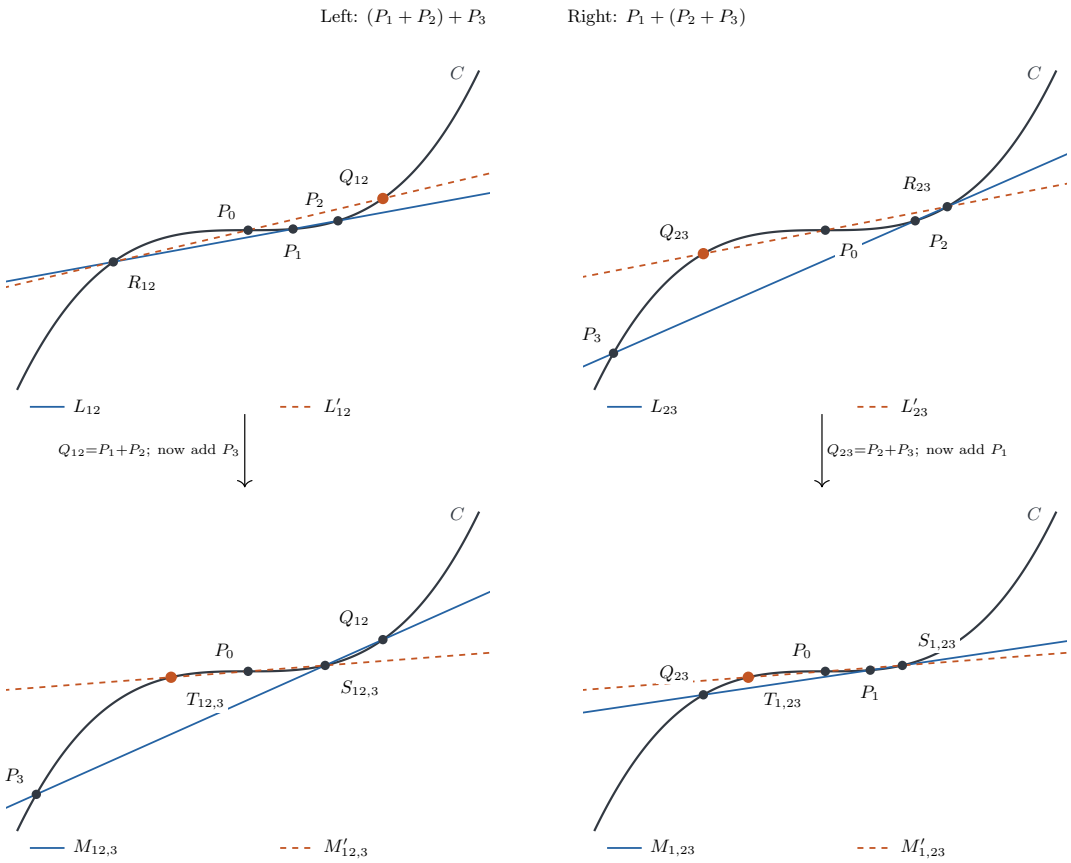
$$L = L_{12}, \quad L' = L'_{23},$$

and

$$\begin{aligned} PP' &= M_{1,23}, \quad & P'' &= S_{1,23}, \\ QQ' &= L_{23}, & Q'' &= P_3, \\ RR' &= L'_{12}, & R'' &= Q_{12}, \end{aligned}$$

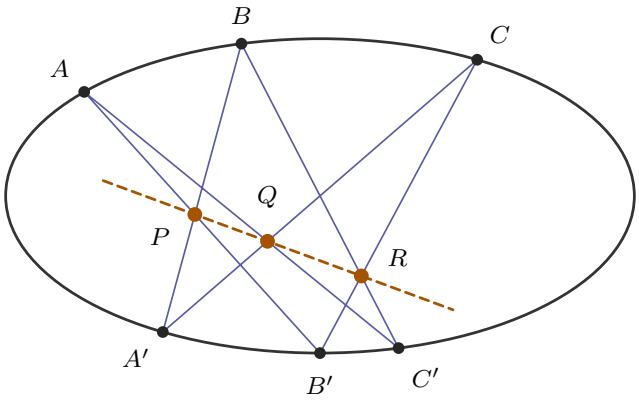
shows $S_{1,23}$, P_3, and Q_{12} are collinear. Therefore, $S_{1,23} = S_{12,3}$, and so $T_{1,23} = T_{12,3}$.

□



$T_{12,3} = (P_1 + P_2) + P_3 = P_1 + (P_2 + P_3) = T_{1,23}$.

5. Prove Pascal's theorem: if A, B, C, A', B', C' are any six points on a conic, then the points $P = AB'.A'B$, $Q = AC'.A'C$, and $R = BC'.B'C$ are collinear.



Proof. Denote the conic by C, and let L be the line PQ. By (4.5), any cubic curve through the points $A, B, C, A', B', C', P, Q$ passes through a uniquely determined point, which we claim to be R. Indeed, the divisors $AB' + A'C + BC'$ and $AC' + A'B + B'C$ intersect at R, so $C + L$ passes through R. Hence, L passes through R. □

6. Generalize (4.5) as follows: given 13 points $P_1, \ldots, P_{13}$ in the plane, there are three additional determined points P_{14}, P_{15}, P_{16}, such that all quartic curves through $P_1, \ldots, P_{13}$ also pass through P_{14}, P_{15}, P_{16}. What hypotheses are necessary on $P_1, \ldots, P_{13}$ for this to be true?

7. If D is any divisor of degree d on the cubic surface (4.7.3), show that

$$p_a(D) \leq \begin{cases} \frac{1}{6}(d-1)(d-2) & \text{if } d \equiv 1, 2 \mod 3 \\ \frac{1}{6}(d-1)(d-2) + \frac{2}{3} & \text{if } d \equiv 0 \mod 3. \end{cases}$$

Show furthermore that for every $d > 0$, this maximum is achieved by some irreducible nonsingular curve.

Proof. Let $D = a\ell - \sum b_i e_i$ be a divisor of degree d. By (4.8), we have

$$d = 3a - \sum b_i,$$
$$p_a(D) = \frac{1}{2}(a-1)(a-2) - \frac{1}{2}\sum b_i(b_i - 1),$$

so to find an upper bound on $p_a(D)$, we want to maximize a and minimize b_i. If $d \equiv 0 \mod 3$, then we can take $a = d/3$ and $b_i = 0$ for all i. □

8. Show that a divisor class D on the cubic surface contains an irreducible curve $\iff$ it contains an irreducible nonsingular curve $\iff$ it is either (a) one of the 27 lines, or (b) a conic (meaning a curve of degree 2) with $D^2 = 0$, or (c) $D.L \geq 0$ for every line L, and $D^2 > 0$.

9. If C is an irreducible non-singular curve of degree d on the cubic surface, and if the genus $g > 0$, then

$$g \geq \begin{cases} \frac{1}{2}(d-6) & \text{if } d \text{ is even, } d \geq 8, \\ \frac{1}{2}(d-5) & \text{if } d \text{ is odd, } d \geq 13, \end{cases}$$

and this minimum value of $g > 0$ is achieved for each d in the range given.

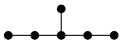
10. A curious consequence of the implications (iv) $\implies$ (iii) of (4.11) is the following numerical fact: Given integers $a, b_1, \ldots, b_6$ such that $b_i > 0$ for each i, $a - b_i - b_j > 0$ for each i, j, and $2a - \sum_{i \neq j} b_i > 0$ for each j, we must necessarily have $a^2 - \sum b_i^2 > 0$. Prove this directly (for $a, b_1, \ldots, b_6 \in \mathbb{R}$) using methods of freshman calculus.

11. *The Weyl Groups.* Given any diagram consisting of points and line segments joining some of them, we define an abstract group, given by the generators and relations, as follows: each point represents a generator x_i. The relations are $x_i^2 = 1$ for each i; $(x_i x_j)^2 = 1$ if i and j are not joined by a line segment, and $(x_i x_j)^3 = 1$ if i and j are joined by a line segment.

(a) The *Weyl group* $\mathbf{A}_n$ is defined using the diagram

of $n-1$ points, each joined to the next. Show that it is isomorphic to the symmetric group Σ_n as follows: map the generators of $\mathbf{A}_n$ to the elements $(12), (23), \ldots, (n-1, n)$ of Σ_n, to get a surjective homomorphism $\mathbf{A}_n \to \Sigma_n$. Then estimate the number of elements of $\mathbf{A}_n$ to show in fact it is an isomorphism.

(b) The *Weyl group* $\mathbf{E}_6$ is defined using the diagram



Call the generators $x_1, \ldots, x_5$ and y. Show that one obtains a surjective homomorphism $\mathbf{E}_6 \to G$, the group of automorphisms of the configuration of 27 lines (4.10.1), by sending $x_1, \ldots, x_5$ to the permutations $(12), (23), \ldots, (56)$ of the E_i, respectively, and y to the element associated with the quadratic transformation based at P_1, P_2, P_3.

(c) Estimate the number of elements in $\mathbf{E}_6$, and thus conclude that $\mathbf{E}_6 \cong G$.

12. Use (4.11) to show that if D is any ample divisor on the cubic surface X, then $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).

Proof. By (4.11), D is ample if and only if D is very ample, so by Bertini's theorem, we can assume D is irreducible and nonsingular as a closed subvariety of X. In particular, $\mathcal{O}_X(-D)$ has no global sections, and we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0 \ ,$$

where $\mathcal{O}_D$ is the structure sheaf of D as a variety. Since X is a hypersurface of $\mathbb{P}^3$, we have

$$\mathrm{H}^1(X, \mathcal{O}_X) = \mathrm{H}^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = 0,$$

so the long exact sequence in cohomology gives

$$0 \longrightarrow \mathrm{H}^0(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^0(D, \mathcal{O}_D) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X(-D)) \longrightarrow 0 \ ,$$

and since D is irreducible, we have

$$h^0(X, \mathcal{O}_X) = h^0(D, \mathcal{O}_D) = 1.$$

Hence by exactness, we have $\mathrm{H}^1(X, \mathcal{O}_X(-D)) = 0$. □

13. Let X be the Del Pezzo surface of degree 4 in $\mathbb{P}^4$ obtained by blowing up 5 points of $\mathbb{P}^2$ (4.7).

(a) Show that X contains 16 lines.

(b) Show that X is a complete intersection of two quadric hypersurfaces in $\mathbb{P}^4$ (the converse follows from (4.7.1)).

14. Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbb{P}^3$ with $d = 8$, $g = 6, 7$; $d = 9$, $g = 7, 8, 9$; $d = 10$, $g = 8, 9, 10, 11$. Combining with (IV, §6), this completes the determination of all possible g for curves of degree $d \geq 10$ in $\mathbb{P}^3$.

15. Let $P_1, \ldots, P_4$ be a fintie set of (ordinary) points of $\mathbb{P}^2$, no 3 collinear. We define an *admissible transformation* to be a quadratic transformation (4.2.3) centered at some three of the P_i (call them P_1, P_2, P_3). This gives a new $\mathbb{P}^2$, and a new set of r points, namely Q_1, Q_2, Q_3, and the images of $P_4, \ldots, P_r$. We say that $P_1, \ldots, P_r$ are in *general position* if no three are collinear, and furthermore after any finite sequence of admissible transformations, the new set of r points also have no three collinear.

(a) A set of 6 points is in general position if and only if no three are collinear and not all six lie on a conic.

(b) If $P_1, \ldots, P_r$ are in general position, then the r points obtained by any finite sequence of admissible transformations are also in general position.

(c) Assume the ground field k is uncountable. Then given $P_1, \ldots, P_r$ in general position, there is a dense $V \subseteq \mathbb{P}^2$ such that for any $P_{r+1} \in V$, $P_1, \ldots, P_{r+1}$ will be in general position.

(d) Now take $P_1, \ldots, P_r \in \mathbb{P}^2$ in general position, and let X be the surface obtained by blowing up $P_1, \ldots, P_r$. If $r = 7$, show that X has exactly 56 irreducible nonsingular curves C with $g = 0$, $C^2 = -1$, and that these are the only irreducible curves with negative self-intersection. Ditto for $r = 8$, the number being 240.

(e) For $r = 9$, show that the surface X defined in (d) has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$.

16. For the *Fermat cubic surface.* $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface?